

LAOAG, Philippines

WMO: 982230

Lat: 18.1832N

Lon: 120.5347E

Elev: 8

StdP: 101.23

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	18.4	19.5	14.9	10.6	22.1	15.8	11.2	22.7	8.5	27.0	7.8	27.0	0.7	90	0.345

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	8.2	34.6	27.8	34.1	27.6	33.5	27.4	29.1	33.4	28.7	32.9	28.3	32.5	4.0	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
27.9	24.1	32.5	27.4	23.3	31.9	27.1	22.9	31.6	94.9	33.3	93.1	33.2	91.5	32.6	31.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.1	7.1	6.4	15.9	36.7	1.7	1.2	14.6	37.6	13.6	38.3	12.7	39.0	11.4	39.9		
			14.6	29.8	1.3	0.5	13.7	30.2	13.0	30.5	12.3	30.8	11.4	31.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	27.6	25.4	26.0	27.3	28.7	29.2	28.8	28.3	27.9	28.0	27.9	27.5	26.4	(6)
(7)		DBStd	1.64	1.49	1.38	1.27	1.00	1.09	1.06	1.06	1.16	0.96	1.12	1.15	1.40	(7)
(8)		HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)
(10)		CDD10.0	6434	477	447	537	560	596	566	568	556	539	554	526	509	(10)
(11)		CDD18.3	3393	219	214	278	310	338	316	309	298	289	296	276	251	(11)
(12)		CDH23.3	36099	1817	1913	2947	3821	4241	3734	3371	3091	3018	3133	2791	2222	(12)
(13)		CDH26.7	14504	634	726	1254	1782	2002	1622	1288	1066	1104	1200	1046	781	(13)
(14)	Wind	WSAvg	2.8	2.9	2.8	2.8	2.6	2.6	2.9	2.8	3.0	2.5	3.0	2.9	3.2	(14)
(15)	Precipitation	PrecAvg	2219	3	2	2	16	136	330	469	539	447	144	45	12	(15)
(16)		PrecMax	3193	24	18	23	201	405	1135	1505	1064	1153	860	239	120	(16)
(17)		PrecMin	1394	0	0	0	0	0	26	26	76	40	0	0	0	(17)
(18)		PrecStd	493	6	4	6	37	111	252	378	278	266	174	62	24	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.0	33.5	34.5	35.1	35.5	34.8	34.1	33.6	33.7	34.2	34.1	33.5	(19)
(20)			MCWB	24.8	24.9	26.1	27.6	28.2	28.8	28.8	28.6	28.0	26.9	26.3	25.4	(20)
(21)		2%	DB	31.8	32.2	33.2	34.2	34.7	33.9	33.2	32.8	32.8	33.2	33.1	32.3	(21)
(22)			MCWB	24.6	25.0	25.9	27.4	28.3	28.4	28.3	28.1	27.8	26.9	26.1	25.3	(22)
(23)		5%	DB	30.9	31.3	32.6	33.6	34.0	33.1	32.5	32.1	32.1	32.4	32.2	31.6	(23)
(24)			MCWB	24.6	24.9	25.8	27.3	28.0	28.1	27.9	27.7	27.5	26.8	26.0	25.0	(24)
(25)		10%	DB	30.0	30.6	31.8	32.9	33.2	32.6	31.9	31.3	31.4	31.8	31.4	30.7	(25)
(26)			MCWB	24.1	24.6	25.6	27.0	27.7	27.8	27.6	27.4	27.2	26.6	25.9	24.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	26.7	27.1	27.9	28.8	29.5	29.5	29.4	29.1	29.0	28.7	28.2	27.3	(27)
(28)			MCDWB	30.9	31.4	32.7	33.9	34.4	33.8	33.2	32.7	32.7	32.6	32.2	31.5	(28)
(29)		2%	WB	26.0	26.3	27.2	28.2	28.8	28.9	28.7	28.5	28.3	28.0	27.5	26.6	(29)
(30)			MCDWB	30.3	30.7	32.0	33.4	33.7	33.2	32.6	32.0	31.9	31.9	31.5	30.8	(30)
(31)		5%	WB	25.3	25.7	26.7	27.7	28.4	28.5	28.2	28.0	27.8	27.5	26.9	25.9	(31)
(32)			MCDWB	29.6	30.2	31.4	32.7	33.2	32.6	32.0	31.5	31.4	31.3	31.0	30.1	(32)
(33)		10%	WB	24.7	25.1	26.1	27.3	27.9	28.0	27.8	27.5	27.4	26.9	26.4	25.2	(33)
(34)			MCDWB	28.8	29.6	30.7	32.1	32.6	32.1	31.3	30.7	30.8	30.6	30.2	29.3	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	9.7	9.6	9.5	9.0	8.2	7.5	6.9	6.4	7.1	7.9	8.2	8.6	(35)
(36)			MCDBR	10.8	10.7	10.3	9.7	9.2	8.4	8.1	7.8	8.1	9.2	9.7	10.2	(36)
(37)			MCWBR	5.4	5.3	5.1	4.6	4.4	4.1	4.1	3.9	4.0	4.4	4.5	4.9	(37)
(38)		5% WB	MCDWB	9.7	9.7	9.4	9.2	8.8	8.2	7.9	7.4	7.8	8.3	8.6	9.3	(38)
(39)			MCWBR	5.4	5.4	5.1	4.6	4.4	4.2	4.1	3.9	4.1	4.3	4.5	5.0	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.407	0.415	0.480	0.501	0.462	0.459	0.459	0.465	0.462	0.457	0.438	0.435	(40)	
(41)		taud	2.433	2.402	2.222	2.182	2.344	2.366	2.362	2.334	2.333	2.326	2.390	2.366	(41)	
(42)		Ebn at Noon	875	887	838	819	841	835	837	838	842	835	837	833	(42)	
(43)		Edh at Noon	109	118	145	152	128	124	125	129	128	126	113	114	(43)	
(44)	All-Sky Solar Radiation	RadAvg	4.51	5.28	6.04	6.59	5.97	5.77	5.30	4.83	5.03	4.92	4.43	4.01	(44)	
(45)		RadStd	0.22	0.35	0.28	0.31	0.43	0.57	0.58	0.54	0.45	0.37	0.40	0.28	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(46)
		N/A	N/A	N/A	-0.39	-0.33	-0.25	N/A	N/A	N/A	N/A	
(47)	Regional (4 neighbors)	N/A	N/A	N/A	-0.33	-0.24	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page